Demonstration of Proposed Features

Date	15 March 2026
Team ID	SWTID-2026-1940
Project Name	Team Enrollment Confirmation - EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Explanation Module	Concept explanation logic using LaMini-Flan-T5 model for simplified educational responses (/explain endpoint)	Implemented	Yes	Completed
2	QnA Module	General knowledge & academic question answering using Gemini 1.5 Pro API (/qa endpoint)	Implemented	Yes	Completed
3	Quiz Generation Module	Generates 3 MCQs with 4 options and answer verification from passage/topic (/quiz endpoint)	Implemented	Yes	Completed
4	Summarization Module	Summarizes long educational passages into key concise bullet/paragraph points (/summarize endpoint)	Implemented	Yes	Completed
5	Learning Path Module	Adaptive learning recommendations structured from beginner to advanced levels (/learn/recommendations endpoint)	Implemented	Yes	Completed

Feature Implementation Summary:

Total Features Proposed	**5**
Total Features Implemented	**5**
Total Features Demonstrated	**5**

Overall Implementation Rate (%)	**100%**

Team Member Work Allocation Breakdown (Assigned by Name Order)

1. R. Priya dharshini *(Team Lead & Core AI Developer)*

• **Assigned Module:** Explanation Module & Project Architecture

• **Responsibilities:** Implemented `LaMini-Flan-T5-783M` local model integration for concept simplification (`/explain` API endpoint), established overall FastAPI folder architecture, and managed team coordination.

2. S. Shobana *(Generative AI Engineer)*

• **Assigned Module:** QnA Module

• **Responsibilities:** Designed and integrated Google Gemini 1.5 Pro API for context-aware academic question answering (`/qa` API endpoint), handling prompt engineering and general query processing.

3. S. Rakshana *(Assessment & Logic Developer)*

• **Assigned Module:** Quiz Generation Module

• **Responsibilities:** Developed MCQ generation pipeline (`/quiz` API endpoint), implemented structured JSON response parsing (`clean_json_block` function), and handled option distractor validation logic.

4. E. Shylaja *(NLP & Summarization Developer)*

• **Assigned Module:** Summarization Module

• **Responsibilities:** Built the passage summarization engine (`/summarize` API endpoint) using Gemini API, optimized response length for student revision, and implemented error exception handling.

5. S. Sheeba *(Full-Stack & Integration Lead)*

• **Assigned Module:** Learning Path Module & Frontend Interface

• **Responsibilities:** Developed adaptive learning path recommendations (`/learn/recommendations`), constructed UI template (`index.html` / `style.css`), and connected REST endpoints with Uvicorn server deployment.